Question		Answer	Marks	AO	Guidance
12	(a)	$\log M = \log(ak^t)$ $\log M = \log k^t + \log a$ $\log M = t\log k + \log a$	 B1 [1]	 2.1	 This line of working must be seen as well as the given result AG
12	(b)(i)	Intercept $\log a = -0.2403$ $a = 10^{-0.2403} = 0.575$ Gradient $\log k = -0.0214$ $k = 10^{-0.0214} = 0.9519$	M1 A1 A1 [3]	3.1a 3.3 3.3	Equates $\log a$ to the intercept of the given line or equates $\log k$ to the gradient of the given line Accept awrt 0.57 or 0.58 Accept one answer in index form Accept awrt 0.95. Second value must be evaluated
	(b)(ii)	This means that every minute, the mass of salt is reduced to 95% of its mass at the start of the minute	B1 [1]	3.3	Also accept that the mass is reduced by 4.8% or 5% per minute FT their value Accept without a value for "the proportion of the mass of salt preserved each minute" oe
12	(c)	When $t = 0,\ M[= a] = 0.575$ Which is quite close to the given value 0.5 kg	M1 A1 [2]	3.4 3.5a	Uses $t = 0$ and their positive value of a. Also allow for $\log 0.5 = -0.3010$ Compares their M with 0.5 kg or their $\log 0.5$ with -0.2403 (must be seen) Small percentage error so reasonable model. Allow for not very good fit

Question		Answer	Marks	AO	Guidance
12	(d)	$\log 0.01 = -0.0214t - 0.2403$	M1	3.4	Uses $\log 0.01$ in the given equation soi
		$t = 82.2$	A1	3.4	Allow value between $82 \leq t \leq 83$
		Alternative method When $0.01 = ak^t$	M1		Uses their values of a and k to form an equation for t and attempt to solve. May be implied by a value for t seen
		$t = 82.2$	A1		To allow for values of a and k rounded to 2 sf, allow value $78 \leq t \leq 83$
			[2]		
12	(e)	$M = ae^{-ht} = ak^t$ So $e^{-h} = k$	M1	3.1a	Equates expressions for M soi FT their k and a
		$h = -\ln k = 0.0493$ $M = 0.575e^{-0.0493t}$	A1FT	1.1	FT their a only. Allow $0.049 < h < 0.052$ Allow $M = 0.575\, e^{-\ln 0.952 t}$
		Alternative methods			
		Uses given value for initial mass $M = 0.01 = 0.5e^{-ht} = 0.5e^{-82h}$	M1		Allow the use of one of the data points from the table for M1 only
		$h = 0.0477$ so $M = 0.5e^{-0.0477t}$	A1		Allow $0.0471 < h < 0.0502$
		Uses their a as initial mass $M = 0.575e^{-ht}$ $0.01 = 0.575e^{-82h}$	M1		Allow the use of one of the data points from the table for M1 only Allow their value of a
		$h = 0.0494$ so $M = 0.575e^{-0.0494t}$	A1		Allow $0.0488 < h < 0.0519$
			[2]		
12	(f)	Rate of decrease is hae^{-ht}	M1	1.2	Soi Allow sign error. FT their h (Q. asks for rate of *decrease*)
		At $t = 10$ rate of decrease is $0.0493 \times 0.575e^{-0.0493\times 10} = 0.0173$ ... [kg per minute]	A1	3.1a	Accept values between 0.017 and 0.018
			[2]		